

Types of Computer Viruses

1. **Boot Sector Virus**
 - Attacks boot sector of hard disk
 - Activates when system starts
2. **File Virus (Program Virus)**
 - Infects executable files (.exe, .com)
3. **Macro Virus**
 - Infects documents (Word, Excel) using macros
4. **Multipartite Virus**
 - Infects both boot sector and files
5. **Polymorphic Virus**
 - Changes its code to avoid detection
6. **Resident Virus**
 - Stays in memory and infects files automatically
7. **Direct Action Virus**
 - Activates only when infected file is executed
8. **Overwrite Virus**
 - Overwrites file content, destroying data

Discuss Various Attacks on Digital Signatures

1. Definition:

Digital Signature is a cryptographic technique used to verify the authenticity, integrity, and non-repudiation of a message. However, it is vulnerable to various types of attacks.

2. Types of Attacks on Digital Signatures:

a) Key-Only Attack:

- Attacker has access only to the public key of the sender
- Tries to generate a valid signature without knowing the private key
- Least powerful attack

b) Known Message Attack:

- Attacker has access to some message-signature pairs
- Uses this information to forge signatures for new messages

c) Chosen Message Attack:

- Attacker can choose messages and obtain their signatures
- More powerful attack
- Common in practical scenarios

d) Adaptive Chosen Message Attack:

- Attacker chooses messages based on previously obtained signatures
- Most powerful type of attack
- Can break weak signature schemes

e) Replay Attack:

- Attacker captures a valid signed message
- Re-sends it to deceive the receiver
- No modification required

f) Forgery Attack:

- Attacker creates a fake signature for a message
- Goal is to make it appear as if it is from a legitimate sender

Types of forgery:

- Existential Forgery: Any valid message-signature pair is created
- Selective Forgery: Signature for a specific message is forged